

Appendix:

Towards Fair and Accurate Short VAAs: Adaptive Ordering and Imputation in Democratic Recommendation Systems

This appendix provides supplementary material referenced in the main text.

A.1 Additional Tables

Table 1 shows the comparative performance of the three ML models used for adaptive and imputed short VAAs. Results are averaged over all selection strategies and grouped by our two datasets.

Table 1: Results for the different ML methods across all settings with 50% stopping and observed + imputed.

Dataset	ML method	Top 1 Accuracy	Top 3 Accuracy	MAE (avg.)
SB	Ridge regression	69.4 [68.6, 70.2]	63.1 [62.3, 63.9]	2.850 [2.823, 2.876]
	Random forest	68.5 [66.8, 70.0]	60.5 [58.8, 62.2]	3.112 [3.052, 3.170]
	Naive Bayes	65.5 [64.7, 66.3]	59.7 [58.9, 60.6]	5.233 [5.180, 5.290]
HA	Ridge regression	66.8 [66.3, 67.4]	48.3 [47.7, 48.9]	2.915 [2.896, 2.935]
	Naive Bayes	63.5 [62.9, 64.1]	46.3 [45.7, 46.9]	5.206 [5.171, 5.243]

A.2 Additional Figures

Figure 1 shows the cross-party fairness for SB, while HA was reported in the main paper.

Figure 1: The SB Top1-accuracy from the best-performing selection strategy (entropy) vs the short VAA (random strategy) for each party, sorted by the number of people having this party as the ground-truth top result. Both methods with imputation of missing items after 50% stopping.

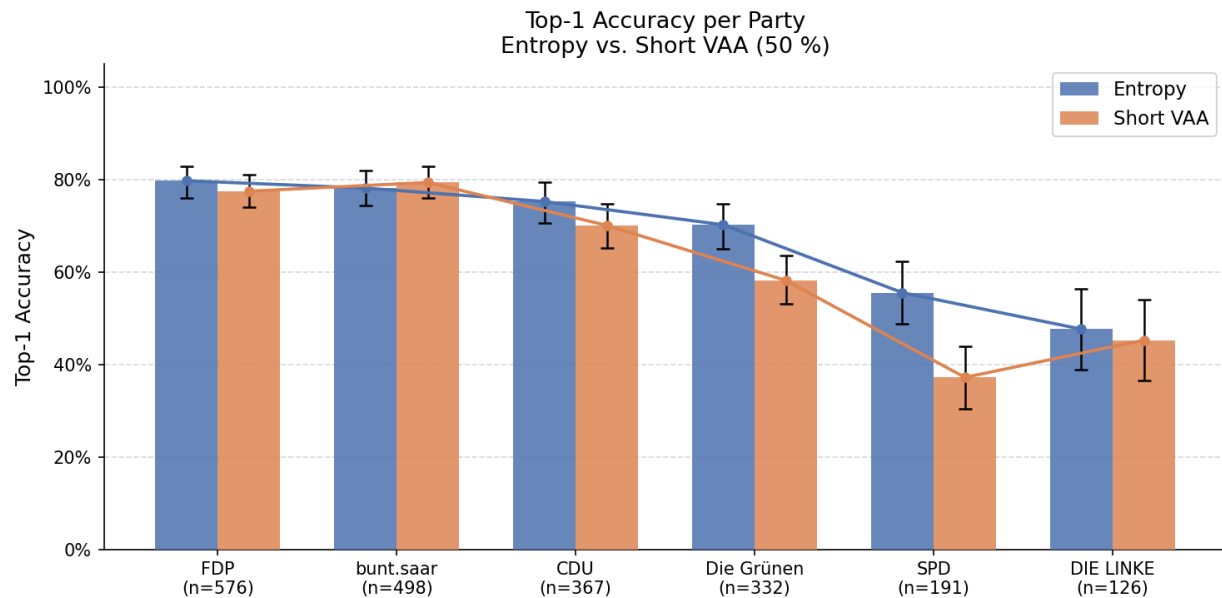


Figure 2 depicts our additional test beyond the typical short VAA stooping criterion ($\approx 50\%$), indicating how many statements we actually need to recover the correct Top1 for each party, based on the HA dataset.

Figure 2: The distribution and median number of items that need to be asked until the correct Top1 is recovered for each party for the Top-1 Change and the random strategies on the HA VAA, additionally grouped by adding imputation or not.

